Project Title

AI-Powered Legal Contract Intelligence Platform

1. Project Overview

Build a web application that allows users to upload legal contracts and interact with them using Artificial Intelligence.

The system should analyze contracts, extract important information, identify risks, compare contracts, and answer user questions using Retrieval-Augmented Generation (RAG).

The objective is to demonstrate practical implementation of:

- Generative AI
- Retrieval-Augmented Generation (RAG)
- Vector Databases
- Large Language Models
- FastAPI
- React

2. Expected Deliverables

By project completion, the student must provide:

Source Code

- Frontend Code
- Backend Code

Documentation

- README.md
- Installation Guide
- Architecture Diagram

Deployment

- Live Demo URL
- GitHub Repository

Presentation

- PPT
- Project Report

3. Functional Requirements

Module 1: User Authentication

Features:

- User Registration
- Login
- Logout

Technology:

- JWT Authentication

Module 2: Contract Upload

User should be able to upload:

- PDF
- DOCX

System should:

- Store file
- Extract text
- Save metadata

Metadata:

- File Name
- Upload Date
- User ID

Module 3: Text Processing

Tasks:

Text Cleaning

Remove:

- Extra spaces
- Invalid characters

Chunking

Split document into chunks

Chunk Size:

- 500 words

Overlap:

- 50 words

Module 4: Embedding Generation

Generate embeddings using:

Option 1:

- OpenAI Embeddings

Option 2:

- BGE-M3

Store embeddings in vector database.

Module 5: Vector Database

Recommended:

Qdrant

Responsibilities:

- Store embeddings
- Similarity search
- Metadata filtering

Module 6: RAG Chat System

User Example:

"What are the payment terms?"

System should:

1. Convert query to embedding
2. Search vector database
3. Retrieve top chunks
4. Send context to LLM
5. Generate answer

Output should include:

- Answer
- Source Chunks

Module 7: Clause Extraction

Automatically extract:

- Effective Date
- Expiry Date
- Payment Terms
- Termination Clause
- Confidentiality Clause
- Governing Law

Store extracted information in PostgreSQL.

Module 8: Risk Assessment

Detect:

High Risk

- Unlimited liability
- Missing termination clause

Medium Risk

- Ambiguous payment terms

- Auto renewal

Low Risk

- Missing governing law

Generate:

Risk Score (0-100)

Example:

Risk Score: 82/100

High Risk:

- Unlimited Liability

Medium Risk:

- Auto Renewal

Module 9: Contract Summarization

Generate:

Executive Summary

50-100 words

Detailed Summary

200-300 words

Module 10: Contract Comparison

User uploads:

Contract A
Contract B

System compares:

- Payment Terms

- Liability
- Termination Conditions
- Renewal Conditions

Generate comparison table.